

100 Gb/s and 2 V V_π InP Mach-Zehnder modulator with an n-i-p-n heterostructure

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An ultra-high bandwidth (BW) and a low V_π InP Mach-Zehnder modulator with an n-i-p-n heterostructure is proposed. The combination of the n-i-p-n heterostructure and the capacitive-loaded travelling-wave electrode provides a modulator with extremely low electrical loss. The device exhibits a 3 dB electro-optic BW of over 67 GHz and a V_π of 2.0 V. A 100 Gb/s non-return-to-zero on-off keying modulation with an extinction ratio of over 10 dB is also realised.

Introduction: An ultra-high-speed and low half-wavelength-voltage (V_π) optical Mach-Zehnder modulator (MZM) is a key component for both coherent systems operating beyond 100 Gb/s and intensity-modulation direct-detection systems operating beyond 100 Gb/s/λ. In general, there is trade-off between electro-optic bandwidth (EO-BW) and V_π because the electro-optic (EO) interaction depends strongly on electrode length which also affects the EO-BW. Thus, previously reported ultra-high speed modulators tend to shorten the electrode length, and an increase in V_π is inevitable [1, 2]. Recently, broadband InP modulators with a low V_π (<2 V) were demonstrated by utilising a low-loss electrical line such as a capacitive-loaded travelling-wave electrode (CL-TWE) [2–5]. However, although these bandwidths (BW) are sufficient for a 32 Gbaud modulation, they are not sufficient for higher baud rate modulations, e.g. over 64 and 100 Gbaud modulations. The main obstacle to higher-speed modulation is the high series resistance of the semiconductor. In particular, the p-doped layer has a contact and bulk resistance about one order of magnitude higher than an n-doped layer. Therefore, the resistance of the p-doped layer must be lower if we are to extend the BW. In this Letter, we report a new high-speed MZM with a low V_π . By employing both an n-i-p-n heterostructure, where a p-doped cladding layer replaces the n-doped layer, [6] and a CL-TWE, we can realise a modulator with an extremely low electrical loss. The device exhibits a 3 dB EO-BW of over 67 GHz and a V_π of 2.0 V with an excess absorption loss of <0.5 dB. Furthermore, we also demonstrated up to 100 Gb/s non-return-to-zero on-off keying (NRZ-OOK) modulations with an extinction ratio (ER) of over 10 dB.

Device structure: Fig. 1 shows a schematic illustration and cross-section diagram of the Mach-Zehnder (MZ) modulation region. An n-contact layer, an n-InP cladding layer, an undoped multi-quantum well (InGaAlAs/InAlAs) core layer, a p-InAlAs electron blocking layer, an n-InP cladding layer, and an n-contact layer were grown from the top on a semi-insulating (SI) InP substrate by using horizontal low-pressure metal-organic vapour-phase epitaxy. The n-doped over-cladding layer in the non-modulation region was replaced by an SI cladding layer to achieve electrical isolation. The MZM has inverted-trapezoidal-shape ridge waveguides in the modulation region and deep ridge waveguides in the I/O passive waveguide region. These waveguides are formed along the [011] stripe direction to obtain a synergistic EO effect [6]. The CL-TWE is then plated on bis-benzocyclobutene. The electrode configuration is based on a conventional series push-pull drive (single-ended 50 Ω design) [1–4]. The total travelling-wave electrode is 3 mm long.

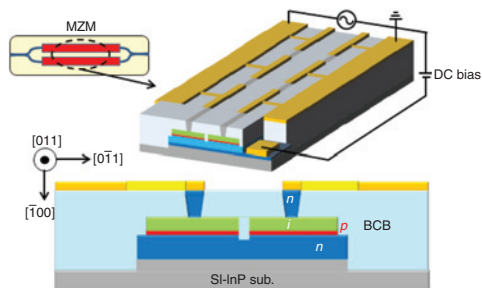


Fig. 1 Schematic illustration and cross-sectional diagram of MZ modulation region

Experimental results: First, we measured the DC ER characteristics for operating wavelengths of 1530, 1540, 1550, and 1560 nm as shown in Fig. 2. A differential voltage was applied across the electrode as with a high-frequency response. The V_π for all the measured wavelengths was set at 2.0 V (DC) by adjusting the DC bias voltage where the excess absorption loss was <0.5 dB. The ER exceeded 22 dB for the entire C-band and was as high as 27 dB for a wavelength of 1560 nm. The entire insertion loss was 10 dB, which included the coupling loss of the lensed fibre (4 dB/facet) and the on-chip loss was estimated to be 2 dB.

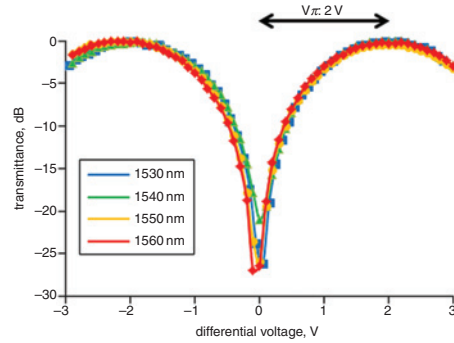


Fig. 2 ER characteristics

We then measured the radio frequency (RF) and EO responses. Fig. 3 shows the small-signal electrical S -parameters of the MZM. The applied DC voltage for a reverse bias was set at −5 V. A 6 dB electrical BW (S_{21}) of over 67 GHz and an electrical reflection (S_{11}) of less than −10 dB were obtained. Furthermore, the measured 3 dB EO-BW (1.5 GHz reference) was over 67 GHz as shown in Fig. 4. The results indicated that the MZM simultaneously satisfied the lightwave-microwave velocity and the characteristic impedance matching conditions.

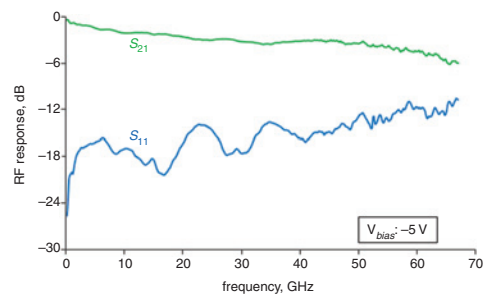


Fig. 3 Small-signal RF responses

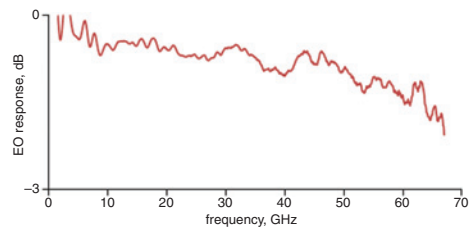


Fig. 4 Small-signal EO response

Finally, we demonstrated the dynamic characteristics. The 50 and 100 Gb/s non-return-to-zero (NRZ) signals with a pseudo-random binary sequence of $2^{31} - 1$ were amplified and fed into a single-RF electrode by an RF probe. The operating wavelength, input optical power, and driving voltage were 1550 nm, +13 dBm, and 2.3 V_{pp} , respectively. Fig. 5 shows generated 50 and 100 Gb/s NRZ-OOK eye diagrams. Clear eye openings and dynamic ERs of over 18 and 10 dB were obtained, respectively. These results indicate that our MZM is superior to other high-speed modulators such as an electro-absorption modulator in providing a high-ER and a low optical loss.

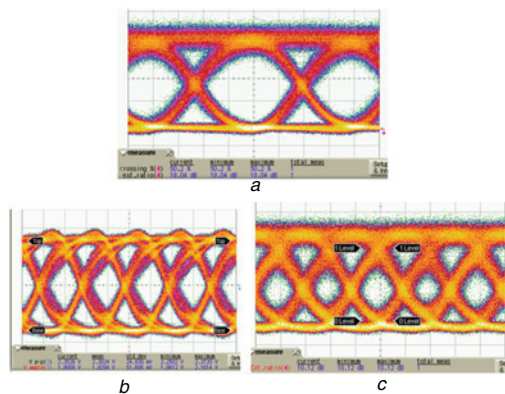


Fig. 5 Measured eye diagrams

a 50 Gb/s modulated signal
b 100 Gb/s electrical input signal
c 100 Gb/s modulated signal

Conclusion: We demonstrated that the combination of an n-i-p-n heterostructure and a CL-TWE can greatly reduce the loss of an RF electrode, which enhances the EO-BW of the MZM compared with a conventional p-i-n heterostructure. A 100 Gb/s NRZ modulation with a dynamic ER of over 10 dB was successfully demonstrated. We believe our low electrical loss structure to be suitable for high capacity transmission systems in various optical networks.

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One or more of the Figures in this Letter are available in colour online.

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